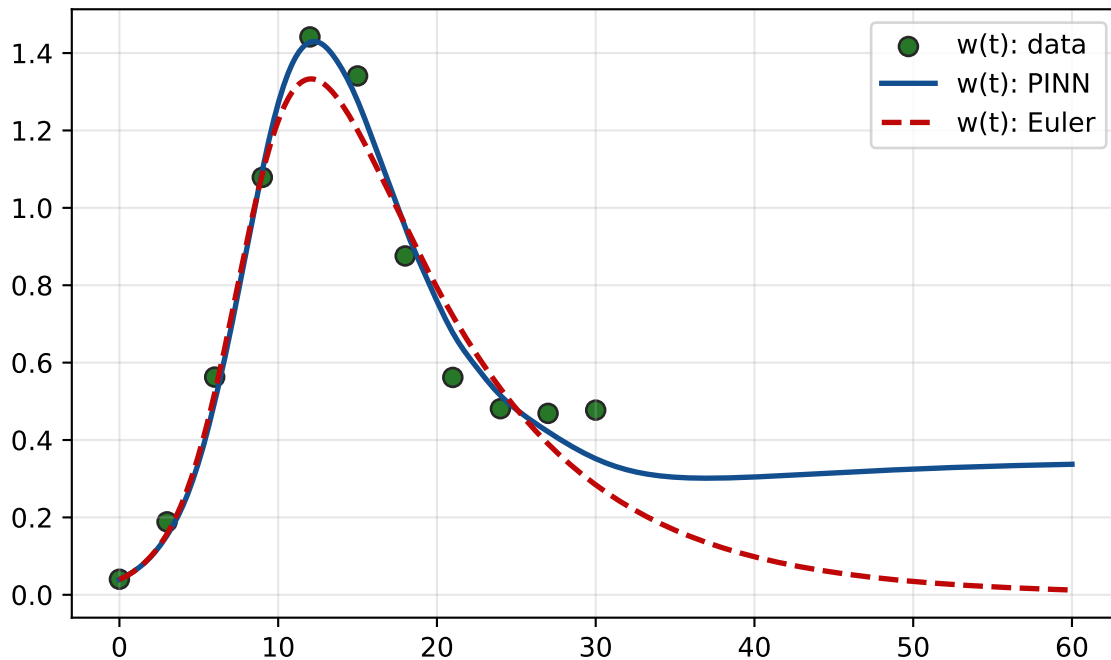
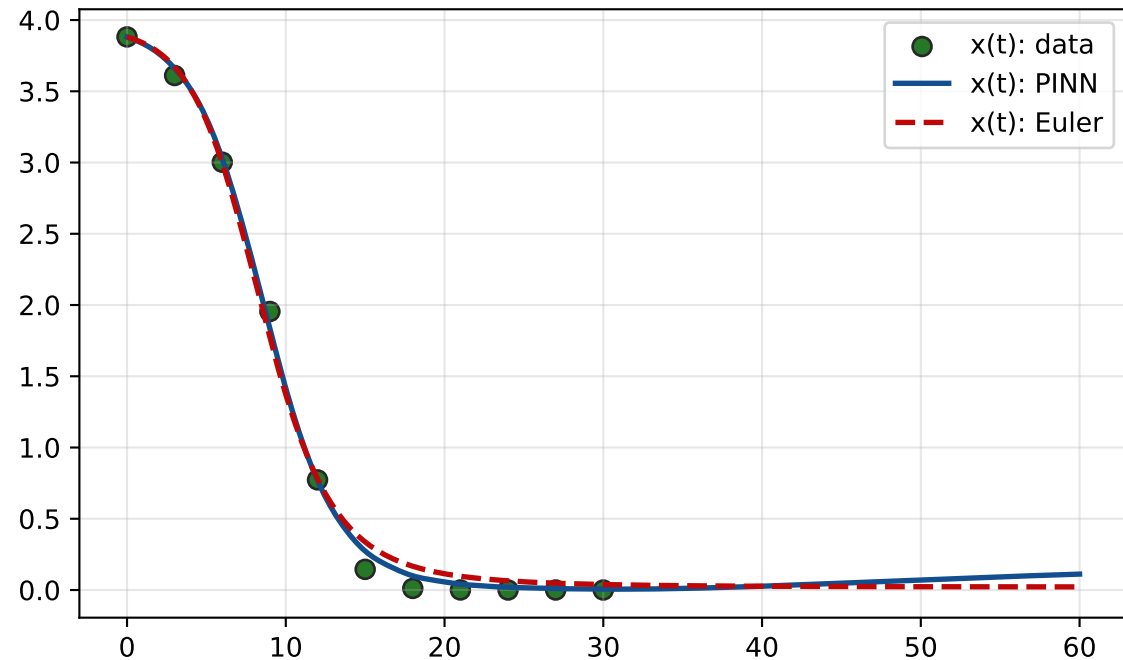


# E2 | PINN vs Euler

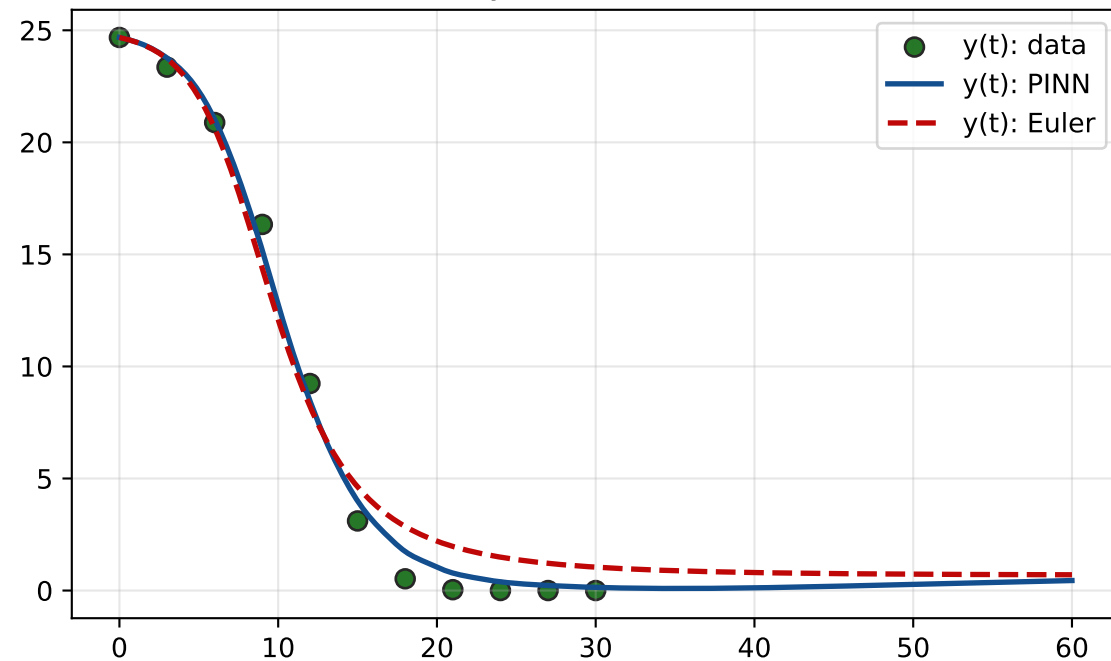
w(t): Biomass



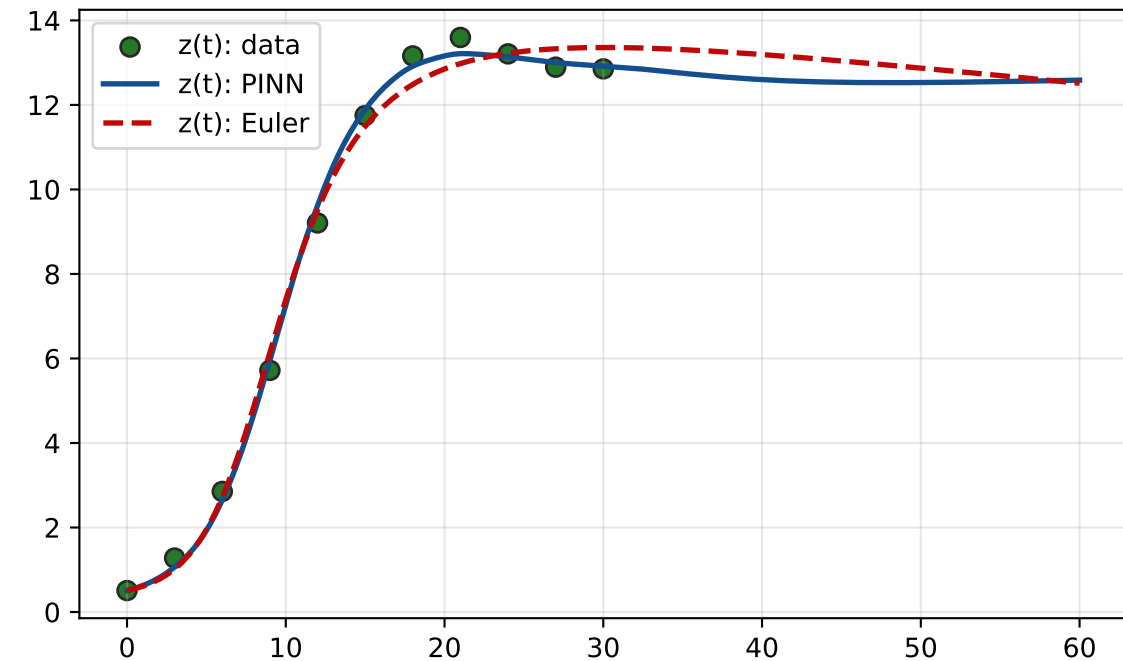
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E2

R2=0.9887, R2\_adj=0.9866, RSS=2.331e+01, AIC=-1.085e+01

cond(JTJ)=1.02e+05, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 1.711979e-02 | 2.217097e-03 | 4.492266e-03 | 1.262752e-02  | 2.161205e-02 | 3.167076e-09 |
| p2    | 8.526950e-02 | 9.918951e-02 | 2.009770e-01 | -1.157075e-01 | 2.862465e-01 | 3.955110e-01 |
| p3    | 7.597187e-03 | 4.721959e-03 | 9.567598e-03 | -1.970411e-03 | 1.716478e-02 | 1.161376e-01 |
| p4    | 2.176594e-01 | 9.645506e-02 | 1.954365e-01 | 2.222288e-02  | 4.130959e-01 | 3.002950e-02 |
| p6    | 1.485175e-01 | 4.199180e-02 | 8.508347e-02 | 6.343405e-02  | 2.336010e-01 | 1.109829e-03 |
| p8    | 6.929449e-02 | 2.023534e-02 | 4.100069e-02 | 2.829380e-02  | 1.102952e-01 | 1.521087e-03 |
| p9    | 3.032833e-03 | 5.514667e-03 | 1.117378e-02 | -8.140944e-03 | 1.420661e-02 | 5.856533e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 3.916913e-02 |
| x0 (fixed)     | 3.881512e+00 |
| y0 (fixed)     | 2.468014e+01 |
| z0 (fixed)     | 5.083666e-01 |
| W0 (fixed)     | 0.000000e+00 |